TabuLLM: AI Benchmark Report

1. Executive Summary: Consolidated Audit Results

This report consolidates the results of the TabuLLM data augmentation pipeline. The objective is to audit the fidelity, diversity, and privacy of the synthetic data generated via LLM inference. Below are the averaged performance indicators across all processed datasets in this session.

Audit Indicator	Global Session Mean (± Std)
Synthetic Fidelity (JSD)	93.8 ± 0.60
Correlation Stability	96.2 ± 0.60
Distribution Consistency	96.0 ± 0.45
Entropy Preservation	96.0 ± 0.45
Mean Z Score Drift	0.0 ± 0.01
Categorical Fidelity	0.0 ± 0.00
Privacy Risk	2.0 ± 0.89
Data Density	100.0 ± 0.00
Coverage Score	81.0 ± 2.76
Feature Accuracy	97.3 ± 0.78
Novelty Score	0.8 ± 0.01

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Dataset Audit: indian_diabetes.csv

1. Dataset Portrait

Attribute	Value
Rows	768
Columns	9
Numeric features	9
Categorical features	0
Missing cells (%)	0.0%

2. Generation Strategy & LLM Config [1]

Parameter	Configuration
Algorithm (*12)	gpt-4o-mini
Folds (*13)	10
Context % (*14)	0%
Generation % (*15)	100%
Join context (*16)	False

3. Statistical Significance & Data Drift

Feature	Orig Mean	Aug Mean	Shift	P-Value	Status
Pregnancies	3.845	3.793	-0.052	1.3%	OK
Glucose	120.895	120.671	-0.224	0.2%	OK
BloodPressure	69.105	68.567	-0.539	0.8%	OK
SkinThickness	20.536	20.070	-0.466	2.3%	OK
Insulin	79.799	76.217	-3.583	4.5%	OK
BMI	31.993	31.661	-0.332	1.0%	OK
DiabetesPedigreeFunction	0.472	0.477	+0.005	1.1%	OK
Age	33.241	32.669	-0.572	1.7%	OK
Outcome	0.349	0.391	+0.042	12.0%	Warn

4. Quality & Fidelity Metrics (Mean ± Std)

Metric Category	Aggregated Score
Synthetic Fidelity (JSD) (*1)	93.8 ± 0.60
Correlation Stability (*2)	96.2 ± 0.60
Distribution Consistency (*3)	96.0 ± 0.45
Entropy Preservation (*4)	96.0 ± 0.45
Mean Z Score Drift (*5)	0.0 ± 0.01
Categorical Fidelity (*6)	0.0 ± 0.00
Privacy Risk (*7)	2.0 ± 0.89
Data Density (*8)	100.0 ± 0.00
Coverage Score (*9)	81.0 ± 2.76
Feature Accuracy (*10)	97.3 ± 0.78

Novelty Score (*11)	0.8 ± 0.01

[1] (*12) Algorithm: Contextual constraint or model parameter.

[2] (*13) Folds: Contextual constraint or model parameter.

[3] (*14) Context %: Contextual constraint or model parameter.

[4] (*15) Generation %: Contextual constraint or model parameter.

[5] (*16) Join context: Contextual constraint or model parameter.

[6] Synthetic Fidelity (JSD): Overall distribution similarity.
[7] Correlation Stability: Multivariate relationships preserved.
[8] Distribution Consistency: Median feature similarity.
[9] Entropy Preservation: Information diversity maintained.
[10] Mean Z Score Drift: Standardized distribution shift.
[11] Categorical Fidelity: Category distribution similarity.
[12] Privacy Risk: Synthetic rows too close to originals.
[13] Data Density: Non-missing cells in augmented data.
[14] Coverage Score: Synthetic coverage of original space.
[15] Feature Accuracy: Mean feature similarity.
[16] Novelty Score: Distance ratio to nearest neighbors.

6. Downstream ML Utility: Comparative Analysis

Model: GB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7467532467532467	0.7428571428571429	0.7376623376623377	-0.009090909090909084	
F1	0.6166023694091699	0.6103636548278197	0.6133842932048663	-0.003218076204303877	

Model: XGB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7292207792207792	0.7220779220779221	0.7246753246753247	-0.004545454545454552	
F1	0.592152063889281	0.5861525957036624	0.5850728128177296	-0.007079251071551318	

Model: DT

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7149350649350652	0.6792207792207792	0.7116883116883117	-0.003246753246753214	
F1	0.5807541882028724	0.5439792288318424	0.5751250919965036	-0.005629096206368822	

Model: NB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7636363636363636	0.7454545454545454	0.7571428571428571	-0.006493506493506496	
F1	0.6399852354635025	0.6125320581811893	0.6272709428564074	-0.012714292607094968	

Model: SVM

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7564935064935064	0.7467532467532467	0.7649350649350649	0.008441558441558427	
F1	0.5733680734057942	0.5592335368564926	0.5992123216649695	0.025844248259175272	

Model: AB

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7493506493506492	0.7435064935064934	0.7603896103896103	0.01103896103896106	
F1	0.6076604413342895	0.5976052044628799	0.6301774900033792	0.02251704866908954	

Model: RF

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7577922077922078	0.7474025974025974	0.7571428571428571	-0.00064935064935064	
F1	0.6191457282029253	0.5958423523036508	0.6192398421849055	9.411398198008736e-05	

Model: LR

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.772077922077922	0.7545454545454545	0.7623376623376623	-0.00974025974025974	
F1	0.6262914445612621	0.6112717466497116	0.6175708591908717	-0.00872058537039038	

Model: MLP

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.6844155844155845	0.65	0.6779220779220779	-0.00649350649350649	
F1	0.4467644316103098	0.4540766194421256	0.4591203427788548	0.01235591116854415	

Model: KNN

Metric	Real (Baseline)	Generated	Augmented	Delta (Lift)	Sig.
ACCURACY	0.7545454545454545	0.7298701298701299	0.7331168831168832	-0.02142857142857145	
F1	0.5998259463389416	0.5617433538542768	0.5810869452577653	-0.01873900108117635	

Appendix: Benchmark Data Trace

Legend: [Blue] = Real Context | [Green] = Generated Synthetic | [Yellow] = Real Training (Remaining) | [Grey] = Real Test Baseline

A. Baseline: Original Dataset (Full Record Sample)

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunct	Age	Outcome
1	6.0	148.0	72.0	35.0	0.0	33.6	0.627	50.0	1.0
2	1.0	85.0	66.0	29.0	0.0	26.6	0.351	31.0	0.0
3	8.0	183.0	64.0	0.0	0.0	23.3	0.672	32.0	1.0
4	1.0	89.0	66.0	23.0	94.0	28.1	0.167	21.0	0.0
5	0.0	137.0	40.0	35.0	168.0	43.1	2.288	33.0	1.0
6	5.0	116.0	74.0	0.0	0.0	25.6	0.201	30.0	0.0
7	3.0	78.0	50.0	32.0	88.0	31.0	0.248	26.0	1.0
8	10.0	115.0	0.0	0.0	0.0	35.3	0.134	29.0	0.0
9	2.0	197.0	70.0	45.0	543.0	30.5	0.158	53.0	1.0
10	8.0	125.0	96.0	0.0	0.0	0.0	0.232	54.0	1.0
11	4.0	110.0	92.0	0.0	0.0	37.6	0.191	30.0	0.0
12	10.0	168.0	74.0	0.0	0.0	38.0	0.537	34.0	1.0
13	10.0	139.0	80.0	0.0	0.0	27.1	1.441	57.0	0.0
14	1.0	189.0	60.0	23.0	846.0	30.1	0.398	59.0	1.0
15	5.0	166.0	72.0	19.0	175.0	25.8	0.587	51.0	1.0
16	7.0	100.0	0.0	0.0	0.0	30.0	0.484	32.0	1.0
17	0.0	118.0	84.0	47.0	230.0	45.8	0.551	31.0	1.0
18	7.0	107.0	74.0	0.0	0.0	29.6	0.254	31.0	1.0
19	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0
20	1.0	115.0	70.0	30.0	96.0	34.6	0.529	32.0	1.0

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B.1 Audit Trace: Fold 1 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	1.0	100.0	74.0	21.0	0.0	20.0	0.572	28.0	0.0	1.0
2	0.0	105.0	60.0	25.0	87.0	29.9	0.488	22.0	0.0	1.0
3	5.0	103.0	60.0	0.0	0.0	28.8	0.108	30.0	0.0	1.0
4	2.0	118.0	64.0	18.0	0.0	28.9	0.268	22.0	1.0	1.0
5	5.0	88.0	68.0	31.0	54.0	29.5	0.6	26.0	0.0	1.0
6	1.0	89.0	76.0	29.0	37.0	30.8	0.28	23.0	0.0	1.0
7	8.0	110.0	76.0	0.0	0.0	27.8	0.237	58.0	0.0	nan
8	1.0	90.0	62.0	18.0	59.0	25.1	1.268	25.0	0.0	nan
9	2.0	56.0	56.0	28.0	45.0	24.2	0.332	22.0	0.0	nan
10	5.0	111.0	72.0	28.0	0.0	23.9	0.407	27.0	0.0	nan

B.2 Audit Trace: Fold 2 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	67.0	88.0	0.0	0.0	35.5	0.434	22.0	0.0	2.0
2	1.0	92.0	68.0	19.0	0.0	26.5	0.165	25.0	0.0	2.0
3	2.0	84.0	72.0	19.0	0.0	27.3	0.229	21.0	0.0	2.0
4	2.0	85.0	58.0	16.0	76.0	26.0	0.654	22.0	0.0	2.0
5	6.0	104.0	70.0	18.0	68.0	30.8	0.245	29.0	1.0	2.0
6	7.0	97.0	50.0	27.0	49.0	22.9	0.741	43.0	1.0	2.0
7	0.0	91.0	68.0	32.0	210.0	39.9	0.381	25.0	0.0	nan
8	3.0	173.0	82.0	48.0	465.0	38.4	2.137	25.0	1.0	nan
9	1.0	147.0	94.0	41.0	0.0	49.3	0.358	27.0	1.0	nan
10	2.0	107.0	74.0	30.0	100.0	33.6	0.404	23.0	0.0	nan

B.3 Audit Trace: Fold 3 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	2.0	95.0	60.0	15.0	160.0	30.9	0.561	22.0	0.0	3.0
2	4.0	83.0	74.0	30.0	0.0	32.0	0.471	63.0	0.0	3.0
3	0.0	165.0	76.0	63.0	255.0	40.6	0.718	26.0	0.0	3.0
4	4.0	134.0	82.0	0.0	0.0	39.1	0.536	32.0	1.0	3.0
5	4.0	125.0	68.0	31.0	0.0	35.8	0.757	23.0	1.0	3.0
6	8.0	194.0	64.0	0.0	0.0	35.1	0.388	29.0	0.0	3.0
7	5.0	158.0	84.0	41.0	210.0	39.4	0.395	29.0	1.0	nan
8	1.0	101.0	50.0	15.0	36.0	24.2	0.526	26.0	0.0	nan
9	0.0	109.0	88.0	30.0	0.0	32.5	0.855	38.0	1.0	nan
10	5.0	147.0	75.0	0.0	0.0	29.9	0.434	28.0	0.0	nan

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B.4 Audit Trace: Fold 4 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	0.0	120.0	74.0	18.0	87.0	30.5	0.495	25.0	0.0	4.0
2	6.0	102.0	72.0	32.0	192.0	37.7	0.26	55.0	0.0	4.0
3	1.0	164.0	74.0	18.0	0.0	28.3	0.256	29.0	0.0	4.0
4	2.0	90.0	60.0	0.0	0.0	27.3	0.269	24.0	0.0	4.0
5	3.0	122.0	100.0	41.0	370.0	46.8	0.259	22.0	0.0	4.0
6	0.0	101.0	0.0	0.0	0.0	25.0	0.253	23.0	0.0	4.0
7	5.0	117.0	86.0	30.0	105.0	39.1	0.251	42.0	0.0	nan
8	3.0	74.0	68.0	28.0	45.0	29.7	0.293	23.0	0.0	nan
9	2.0	129.0	0.0	0.0	0.0	38.5	0.304	41.0	0.0	nan
10	1.0	126.0	56.0	29.0	152.0	28.7	0.801	21.0	0.0	nan

B.5 Audit Trace: Fold 5 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	134.0	76.0	31.0	79.0	34.1	0.153	54.0	1.0	5.0
2	0.0	102.0	80.0	40.0	0.0	33.6	0.192	26.0	0.0	5.0
3	0.0	114.0	66.0	36.0	96.0	33.3	0.233	24.0	0.0	5.0
4	3.0	171.0	78.0	48.0	545.0	37.0	0.702	31.0	1.0	5.0
5	5.0	78.0	0.0	0.0	0.0	23.7	0.317	28.0	0.0	5.0
6	0.0	151.0	74.0	0.0	0.0	25.6	0.178	70.0	1.0	5.0
7	13.0	126.0	90.0	0.0	0.0	43.4	0.583	42.0	1.0	nan
8	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
9	1.0	181.0	78.0	42.0	293.0	40.0	1.258	22.0	1.0	nan
10	3.0	78.0	70.0	0.0	0.0	32.5	0.27	39.0	0.0	nan

B.6 Audit Trace: Fold 6 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	8.0	194.0	92.0	33.0	265.0	33.9	0.305	54.0	0.0	6.0
2	2.0	99.0	70.0	17.0	0.0	28.2	0.229	24.0	0.0	6.0
3	7.0	196.0	70.0	0.0	0.0	30.9	0.542	36.0	1.0	6.0
4	0.0	127.0	84.0	37.0	370.0	36.3	0.718	23.0	0.0	6.0
5	1.0	106.0	56.0	18.0	116.0	24.3	0.299	23.0	0.0	6.0
6	9.0	165.0	84.0	0.0	0.0	35.9	0.151	49.0	1.0	6.0
7	0.0	102.0	52.0	0.0	0.0	25.1	0.078	21.0	0.0	nan
8	4.0	76.0	62.0	0.0	0.0	34.0	0.391	25.0	0.0	nan
9	1.0	103.0	30.0	38.0	83.0	43.3	0.183	33.0	0.0	nan
10	1.0	124.0	74.0	36.0	0.0	27.8	0.1	30.0	0.0	nan

B.7 Audit Trace: Fold 7 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	14.0	100.0	68.0	40.0	192.0	40.5	0.64	51.0	1.0	7.0
2	8.0	100.0	86.0	48.0	191.0	38.9	0.171	63.0	0.0	7.0
3	1.0	85.0	0.0	0.0	0.0	23.5	0.237	21.0	0.0	7.0
4	7.0	103.0	60.0	33.0	156.0	33.6	0.926	44.0	1.0	7.0
5	1.0	116.0	70.0	28.0	89.0	33.2	0.466	21.0	0.0	7.0
6	1.0	181.0	60.0	30.0	392.0	32.8	0.655	53.0	1.0	7.0
7	2.0	85.0	65.0	0.0	0.0	39.6	0.93	27.0	0.0	nan
8	2.0	142.0	82.0	18.0	64.0	24.7	0.761	21.0	0.0	nan
9	9.0	102.0	76.0	37.0	0.0	32.9	0.665	46.0	1.0	nan
10	4.0	117.0	64.0	27.0	120.0	33.2	0.23	24.0	0.0	nan

B.8 Audit Trace: Fold 8 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	4.0	122.0	68.0	0.0	0.0	26.8	0.419	37.0	0.0	8.0
2	4.0	146.0	85.0	27.0	156.0	30.1	1.353	63.0	0.0	8.0
3	1.0	125.0	84.0	24.0	110.0	30.1	0.234	21.0	0.0	8.0
4	9.0	112.0	72.0	0.0	0.0	28.4	0.376	43.0	1.0	8.0
5	14.0	163.0	68.0	41.0	342.0	37.2	0.254	38.0	1.0	8.0
6	2.0	110.0	86.0	19.0	100.0	28.4	0.698	27.0	0.0	8.0
7	1.0	119.0	44.0	47.0	63.0	35.5	0.28	25.0	0.0	nan
8	8.0	196.0	76.0	29.0	280.0	37.5	0.605	57.0	1.0	nan
9	1.0	118.0	58.0	36.0	94.0	33.3	0.261	23.0	0.0	nan
10	1.0	116.0	70.0	28.0	0.0	27.4	0.204	21.0	0.0	nan

B.9 Audit Trace: Fold 9 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	6.0	119.0	88.0	0.0	0.0	29.6	0.451	30.0	1.0	9.0
2	5.0	121.0	76.0	15.0	51.0	25.6	0.126	28.0	0.0	9.0
3	2.0	112.0	68.0	32.0	0.0	42.1	0.375	23.0	0.0	9.0
4	4.0	131.0	86.0	36.0	330.0	29.7	1.072	31.0	0.0	9.0
5	0.0	95.0	50.0	30.0	105.0	34.6	0.346	24.0	1.0	9.0
6	0.0	173.0	76.0	43.0	255.0	40.6	1.159	31.0	0.0	9.0
7	8.0	65.0	72.0	23.0	0.0	32.0	0.6	42.0	0.0	nan
8	10.0	75.0	82.0	0.0	0.0	33.3	0.263	38.0	0.0	nan
9	2.0	146.0	70.0	38.0	360.0	28.0	0.337	29.0	1.0	nan
10	5.0	126.0	78.0	27.0	22.0	29.6	0.439	40.0	0.0	nan

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B.10 Audit Trace: Fold 10 | Gen: 100%

#	Pregnancies	Glucose	BloodPressure	SkinThickness	Insulin	BMI	DiabetesPedigreeFunc	Age	Outcome	_fold_id
1	8.0	188.0	70.0	0.0	0.0	43.6	0.34	43.0	1.0	10.0
2	0.0	101.0	65.0	17.0	0.0	19.6	0.572	21.0	0.0	10.0
3	0.0	125.0	84.0	0.0	0.0	39.0	0.412	46.0	0.0	10.0
4	5.0	121.0	80.0	7.0	87.0	31.6	0.122	54.0	0.0	10.0
5	1.0	196.0	0.0	0.0	0.0	27.5	0.652	29.0	1.0	10.0
6	7.0	137.0	90.0	43.0	145.0	43.3	0.647	43.0	0.0	10.0
7	3.0	180.0	64.0	25.0	70.0	34.0	0.271	26.0	0.0	nan
8	3.0	99.0	80.0	11.0	64.0	19.3	0.284	30.0	0.0	nan
9	4.0	148.0	60.0	27.0	318.0	30.9	0.15	29.0	1.0	nan
10	7.0	136.0	90.0	0.0	0.0	29.9	0.21	50.0	0.0	nan